

WEB APPLICATION FOUNDATIONS

Web App Basics

Client: Machine or application that makes a request of a server.

Server: Machine or application that responds to requests from clients.

Database: Organized collection of information accessed by queries.

Static web page: A page with the same content for every user.

Dynamic web app: An app that provides customized content for each user.

Front end: The part of an app the user interacts with, which runs on the client.

Back end: The part of an app that runs on a server and responds to front-end/client requests.

Requests come in four general categories: get data from the server, add new data to the server, edit data on the server, and delete data on the server.

Modern web apps involve both static and dynamic content.

Front Ends and Native Apps

User interface: All the elements a user sees and interacts with, including graphics, typography, controls, etc.

User experience: How an app's design feels. Considerations include where buttons are placed, how a user flows through different screens, and which elements grab the user's eye.

Hypertext Markup Language (HTML): Provides the basic structure and content of a web page.

Cascading Style Sheets (CSS): Styles the colors, fonts, layouts, and overall look and feel of a web page.

JavaScript (JS): Makes a web page interactive.

Responsive design: Styling web content to fit multiple screen widths (e.g., desktop, tablet, and mobile).

Browser-based app: An app that's designed to be accessed through a web browser.

Native app: An app that runs directly on the client's device.

HTML, CSS, and JS work together to create the modern, interactive web experience we've come to expect.

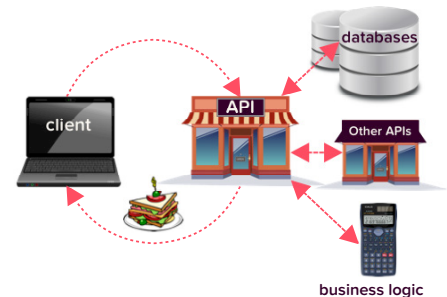
Back Ends and APIs

Application Programming Interface: An interface that connects the front end to back-end resources, ensuring the app's data remains organized and bug-free.

Business logic: Internal ruleset that governs how data flows through the app.

Hypertext Transfer Protocol (HTTP): A procedure for sharing data that both clients and servers can understand.

API endpoint: An address, formatted much like a URL, that returns a resource from a server.



Databases

Relational database: A database with multiple tables that are related to each other via ID numbers.

Structured Query Language (SQL): Computer language used to pose questions to relational databases.

Big data: Data sets so large and complex that they can't be managed with traditional methods.

Non-relational database (NoSQL database): Database that's organized in a more flexible, scalable way in order to deal with the challenges of big data.

Big data is characterized by the "three Vs": volume, variety, and velocity.

Web Frameworks

Web framework: A prebuilt set of tools and resources for building and managing web apps.

Developer experience (DX): The way a developer feels while writing code. Frameworks can help enhance the DX.

UI framework: A set of pre-built elements to help style websites.

JavaScript framework: Pre-built elements to help manage the flow of information to and from the user.

Back-end framework: Tools used on the server side to get an API up and running quickly. Back-end frameworks can be large and complex or lightweight and simple. The latter is sometimes called a **microframework**.

Apps built with a given UI framework tend to look like other apps built using the same framework.

Cyberscurity

Penetration testing: A friendly hacker tries to identify possible entry points into a system then shares any vulnerabilities with system administrators. Penetration testers are also called ethical hackers.

Attack surface: The sum of all the vulnerabilities in a system.

SQL injection: An attack where the hacker uses a text input field to send malicious code to the server for execution. This attack is prevented by sanitizing inputs.

Encryption: The temporary translation of data into an unreadable form.

Brute-force attack: An attempt to crack passwords by having a computer enter every possible combination of keys. This attack is prevented by strong password policies, limiting login attempts, and using *multifactor authentication*.

Multifactor authentication: The practice of requiring more than one proof of a user's identity at login.

Distributed denial of service: An attack where the hacker attempts to overwhelm a server by flooding it with requests from thousands of virtual bots. This attack is prevented by blocking suspicious traffic and increasing server capacity.

Social engineering: An attempt to obtain unauthorized access to systems by manipulating people. This attack is prevented by vigilance and user education.